

Advanced system analysis

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Lecture 5
OO Modelling, **Interaction
Diagrams (Sequence,
Collaboration, & System
Sequence Diagrams)**

LECTURE OBJECTIVES

1. Learn the notation of interaction diagrams.
2. Learn how to visualize system-user interaction using system sequence diagrams, collaboration diagrams, system sequence diagrams.
3. Learn how to decide upon issues such as fork & cascade, or how to implement use-cases.



UML DIAGRAMS

Functional Diagrams (*Use case diagrams*) ✓

- Describe the functional behavior of the system as seen by the user.

Static Diagrams (*Class, Object, & Package diagrams*) ✓

- Describe the static structure of the system: Objects, attributes, associations.

Dynamic Diagrams:

- Interaction diagrams (*Sequence, & Collaboration diagrams*) ⊕

- Describe the interaction between objects of the system.

- **State diagrams** ✗

- Describe the dynamic behavior of an individual object.

- **Activity diagrams** ✓

- Describe the dynamic behavior of a system, in particular the workflow.

UML DYNAMIC MODELLING .. WHY?

UML static modeling involves **use cases**, “**activity diagrams**,” and **class/object modeling**:

- **Use cases** describe what a system should do (*the required behavior/operations*). It did not say which classes should be responsible for which parts of computation
- An initial **Class model** identifies classes and its relationship but do not contain complete list of attributes and operations.
- Thus, a link is required between use cases and class modeling, which leads us to what is known as **dynamic modeling**.
- **Dynamic Modelling** shows how the objects interact by sending messages to implement the required functionality of the software system (to realize a Use-Case).

UML DYNAMIC MODELLING: INTERACTION DIAGRAMS

Interaction diagrams are UML notations representing dynamic modeling that are used to help make and record decisions relating to the behavior (*operations*) defined for each class.

There are two types of interaction diagrams:

1. Sequence Diagram ..
2. Communication (Collaboration) Diagram ..

Both are semantically equivalent to each other, and used for quite similar purposes, but each has different features as described next.

UML INTERACTION DIAGRAMS

SEQUENCE VERSUS COLLABORATION DIAGRAMS

	Sequence Diagrams	Collaboration (Communication) Diagrams
Definition	Shows object interaction arranged in time sequence. That is, a 2-dimensional diagram, horizontally the objects participating in the interaction are depicted, while vertical dimension represents time.	Shows how cooperating objects dynamically interact with each other by sending and receiving messages.
Strengths	Time relationship extremely clear.	Objects' interactions extremely clear.
Weaknesses	Not easy to see the overall pattern of message flow.	Actual sequence of messages is difficult to see.

SEQUENCE DIAGRAMS

A ***sequence diagram*** shows the participants (objects) in an interaction (a use case or part of it) and the sequence of messages among them.

Each Use Case requires one or more sequence diagrams to describe its behaviour.

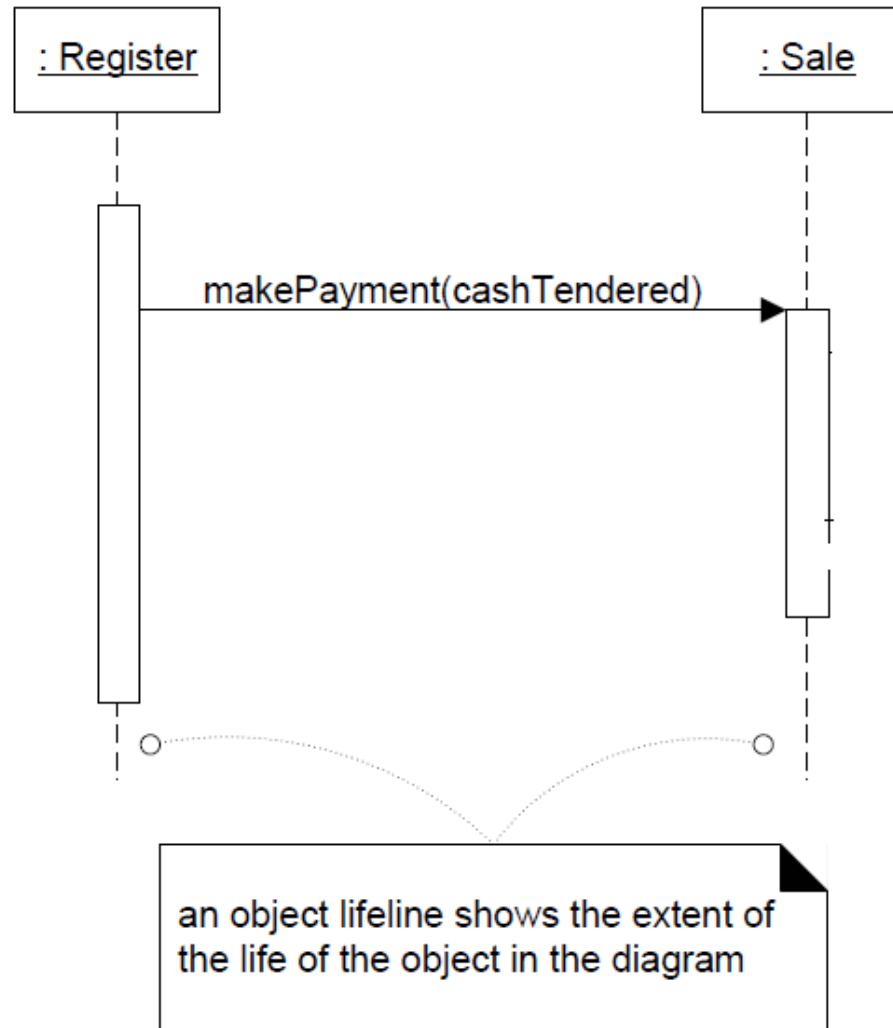


SEQUENCE DIAGRAMS

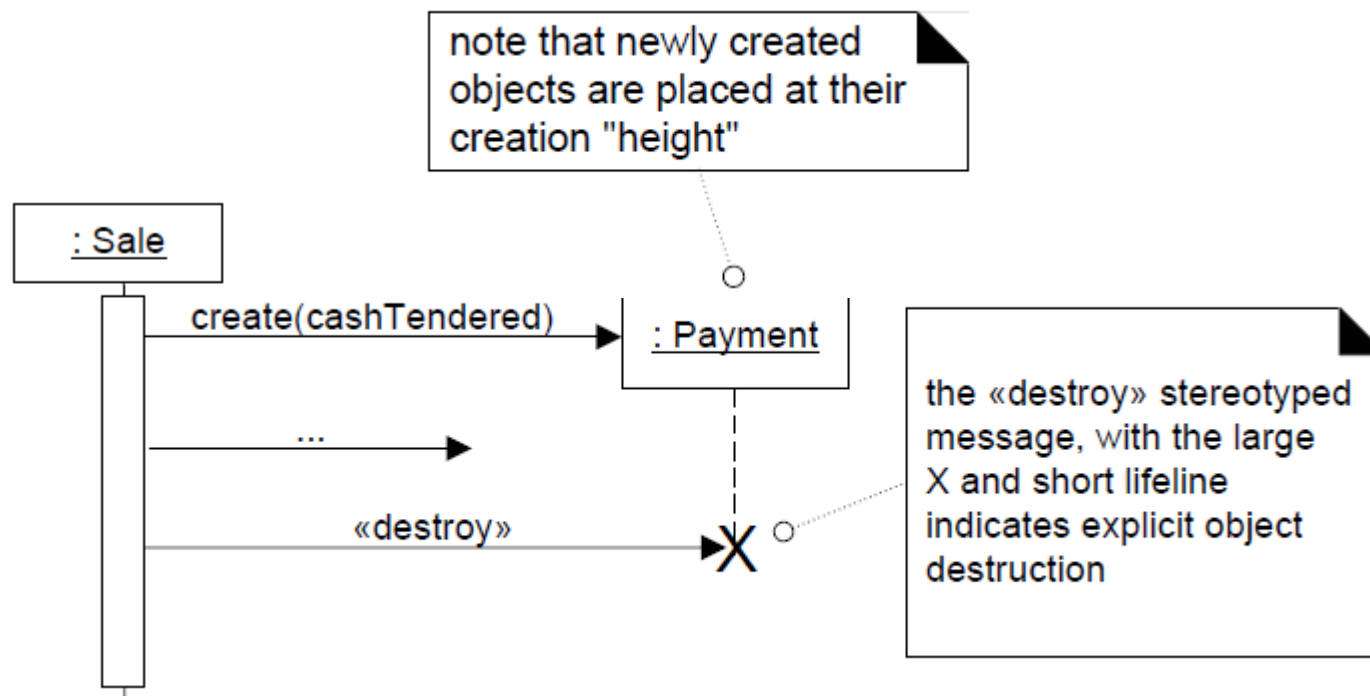
Starting by building a sequence diagram:

- Begin by looking at **interactions implied by use cases**. Consider a **use case scenario** to be modeled.
- You need to know where the **first message** in the use case originates. (*for example, a user interface*).
- You need to consider the **message flow**;
 - To where the initial message should be sent?
 - This implies which class should have this method in its protocol.
- Use the **post-condition(s)** of the use cases which reflect how the system must change.
 - Take into consideration the **main scenario and the alternative ones**.
 - It may be helpful to **use a pair of object diagrams**, showing the states before and after the operation in question.

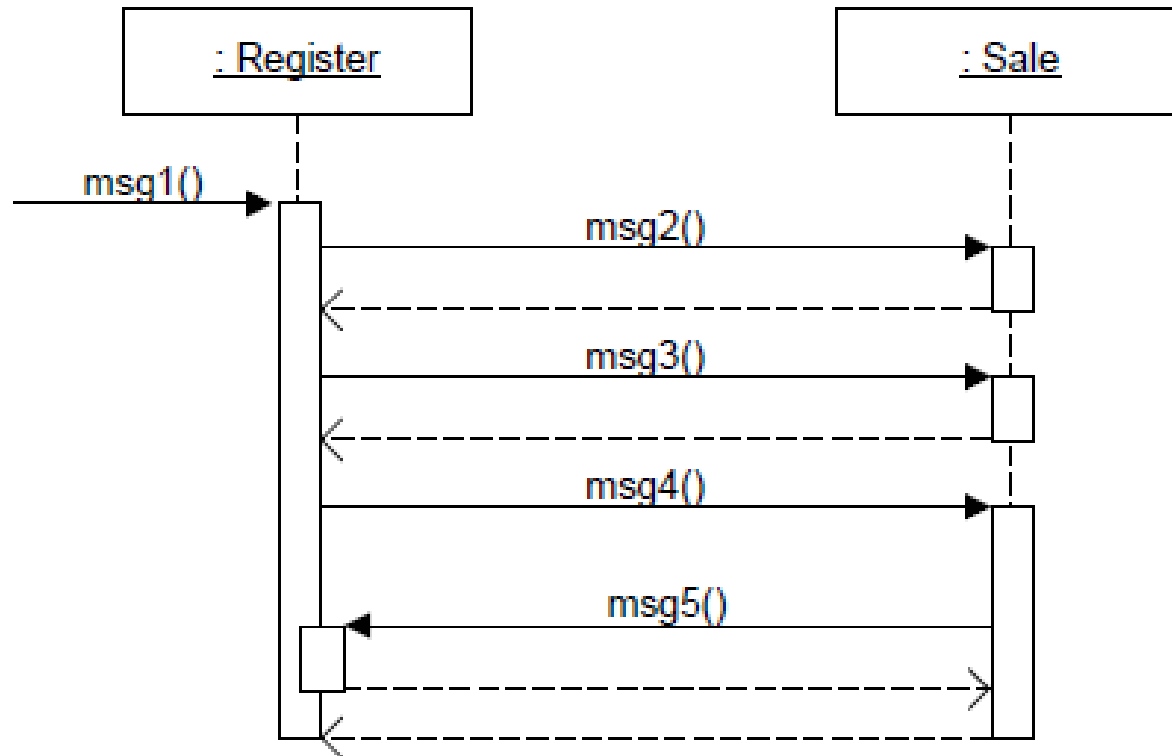
OBJECT LIFELINES



OBJECTS CREATION AND DESTRUCTION



MESSAGES, RETURNS & ACTIVATION BOXES



CONDITIONAL MESSAGES

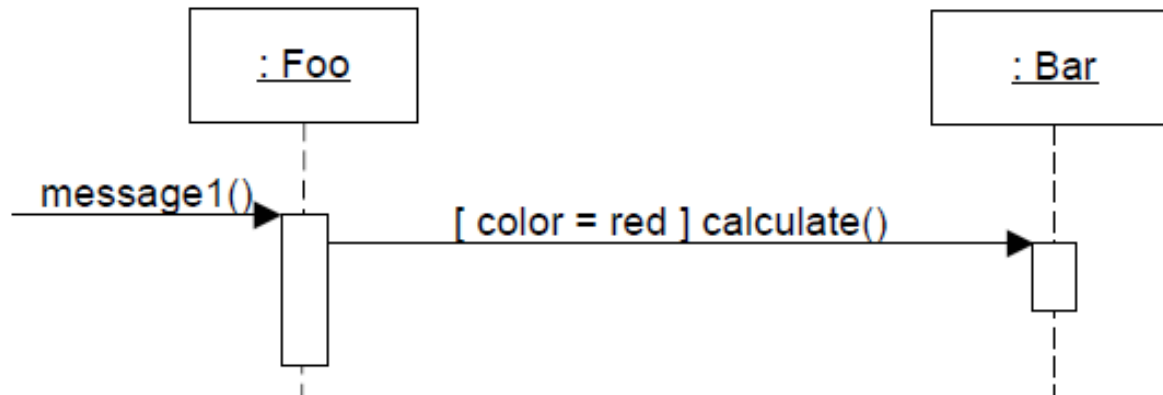
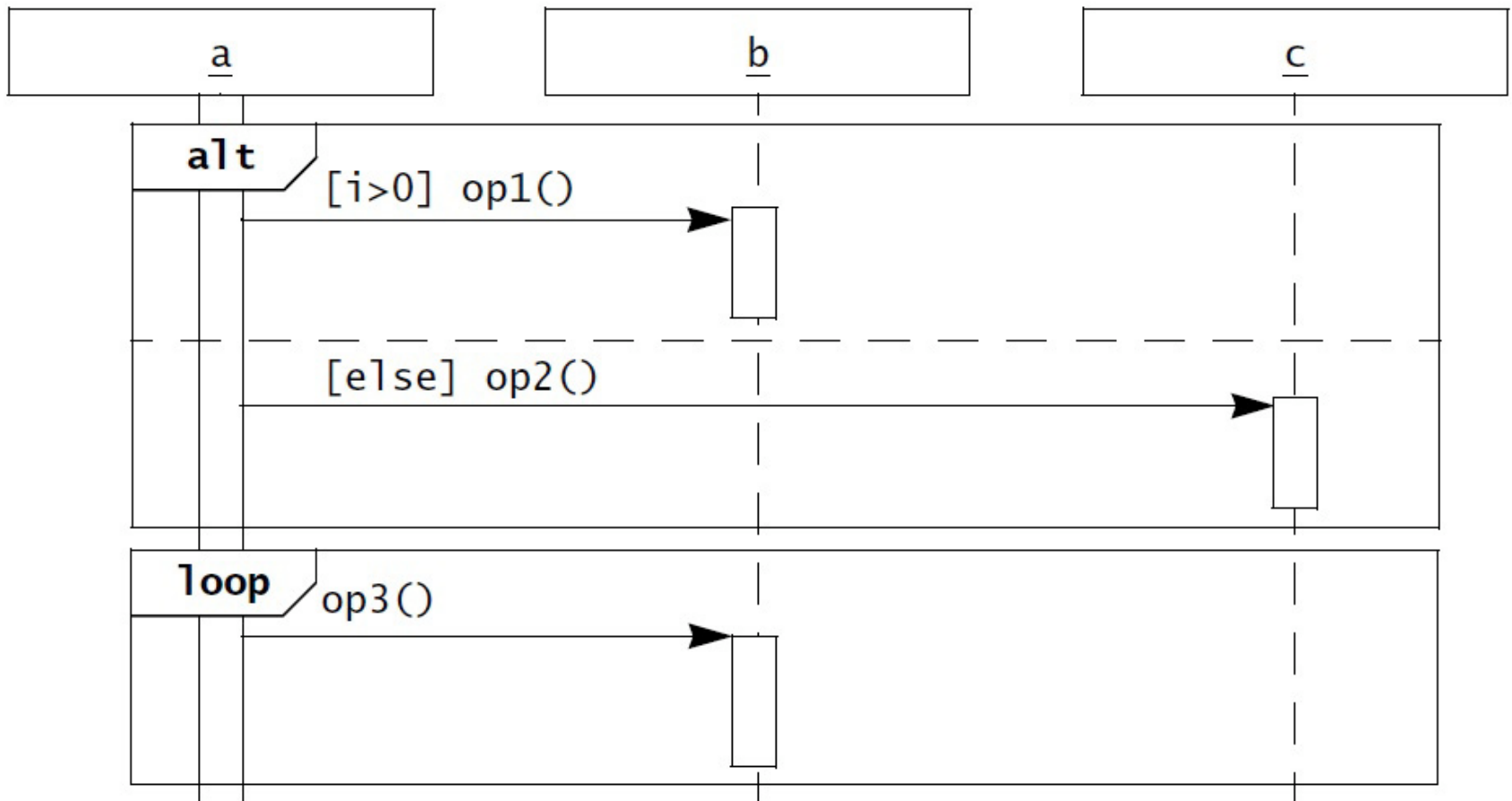
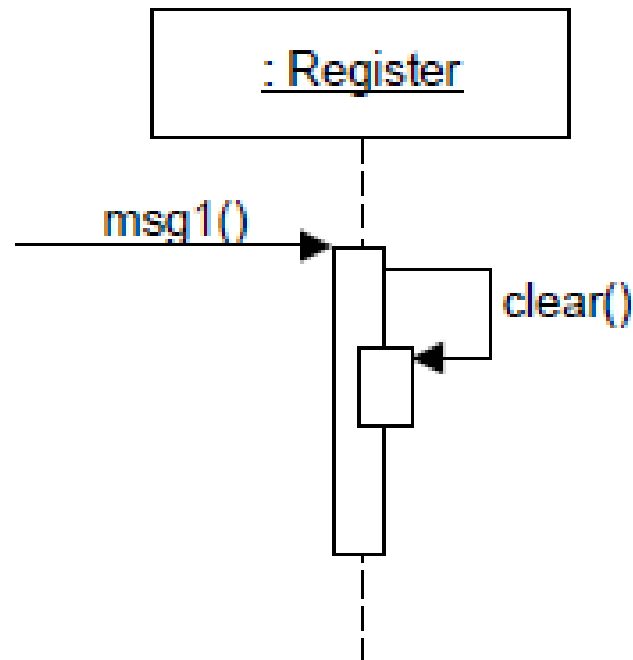


Figure 15.22 A conditional message.

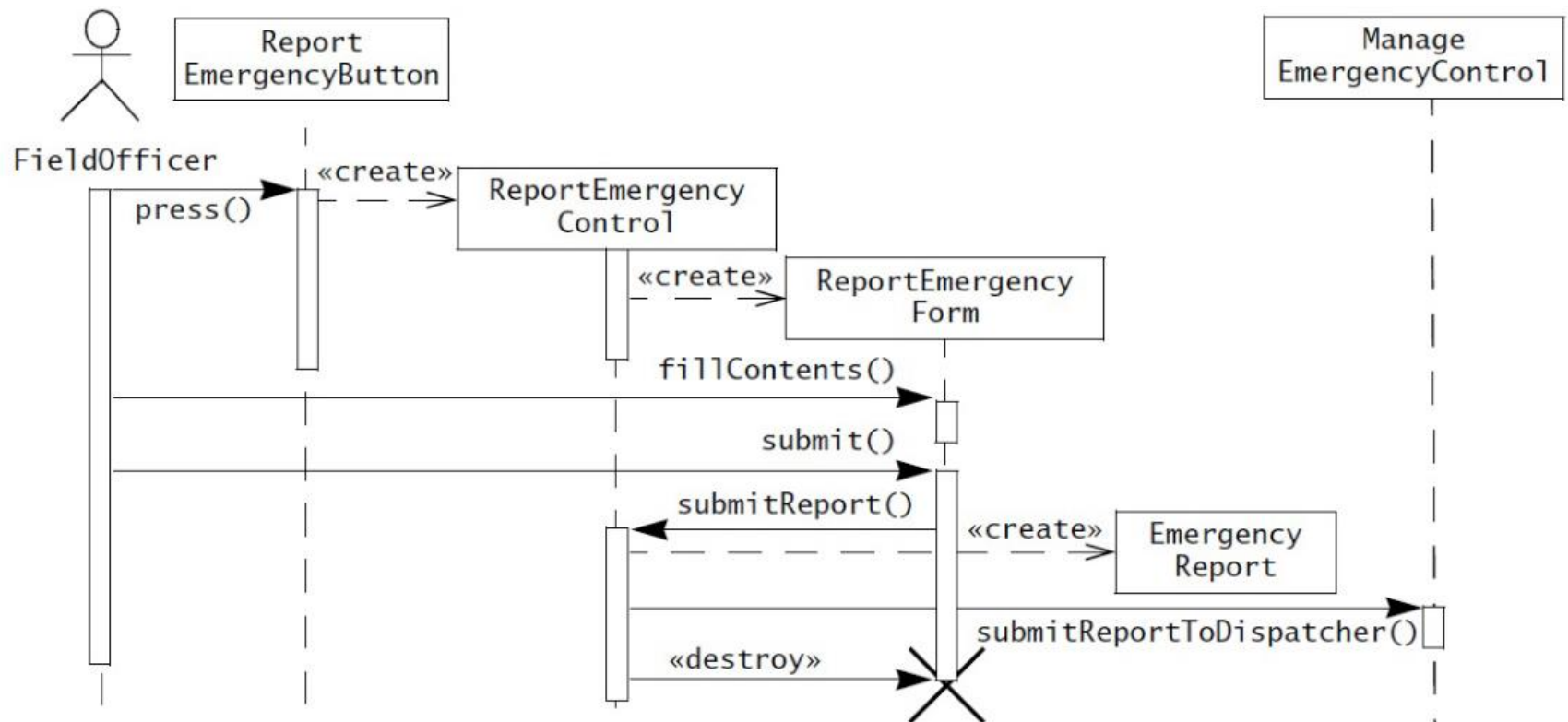
CONDITIONS AND ITERATIONS



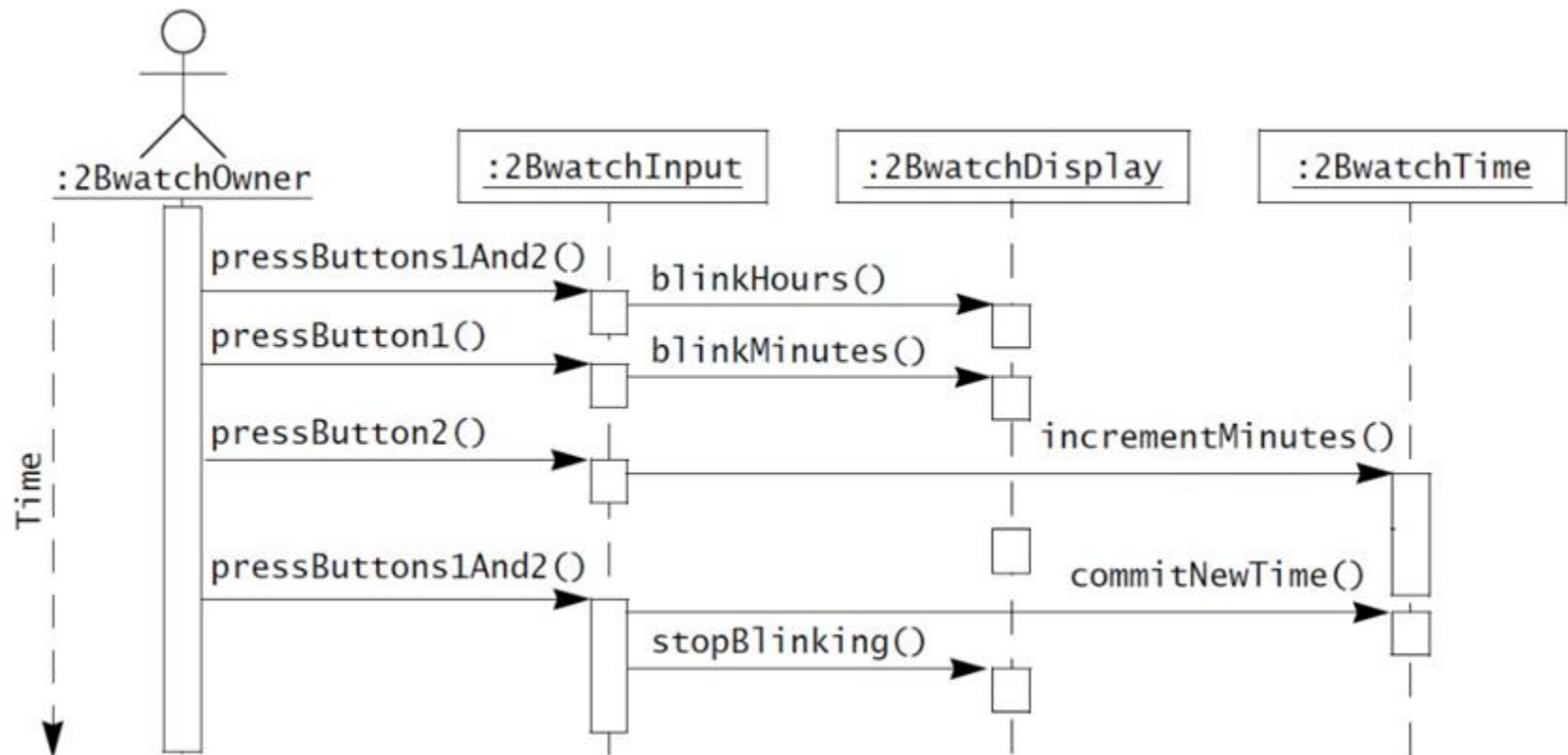
MESSAGES TO "SELF" OR "THIS"



SEQUENCE DIAGRAM: AN EXAMPLE

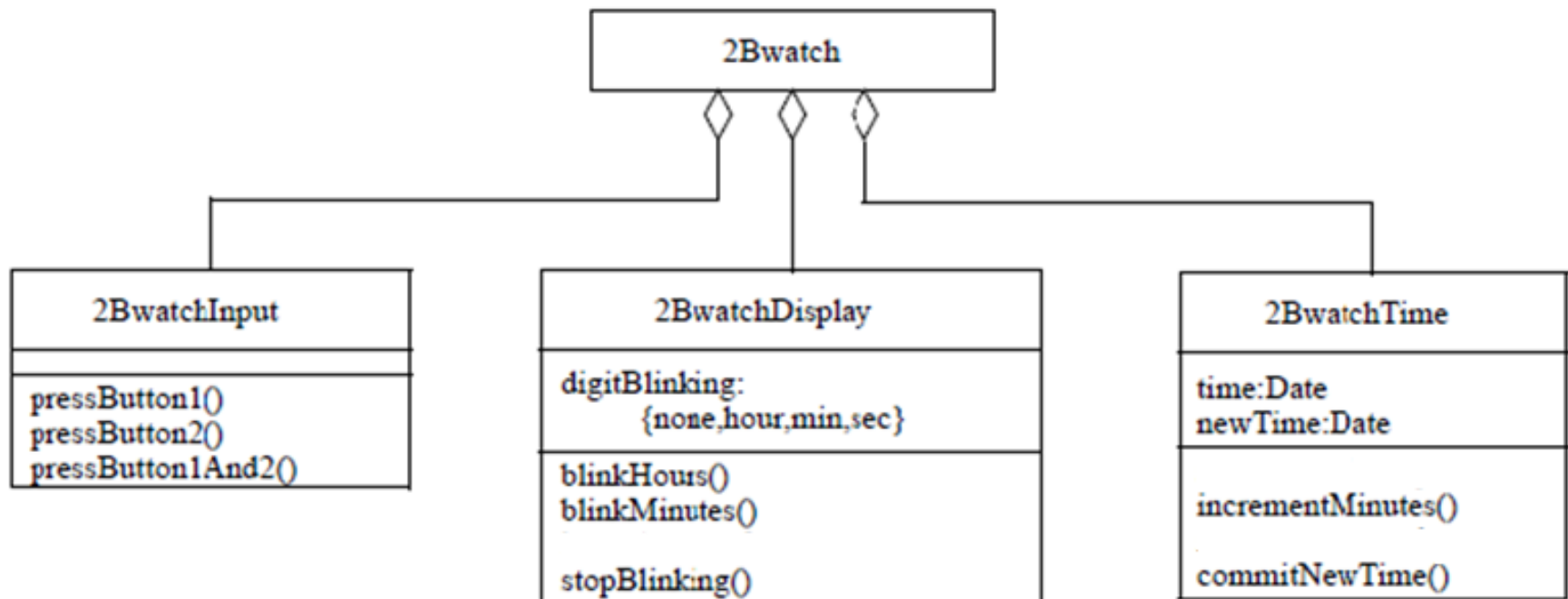


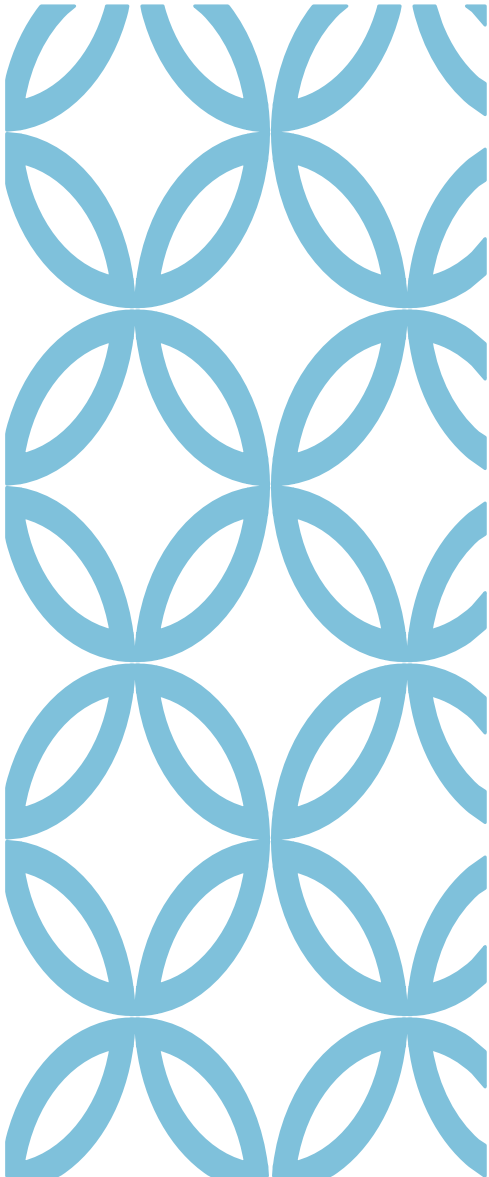
SEQUENCE DIAGRAM: EXAMPLE 2



SEQUENCE DIAGRAM: EXAMPLE 2

(SEQUENCE TO CLASS DIAGRAM)





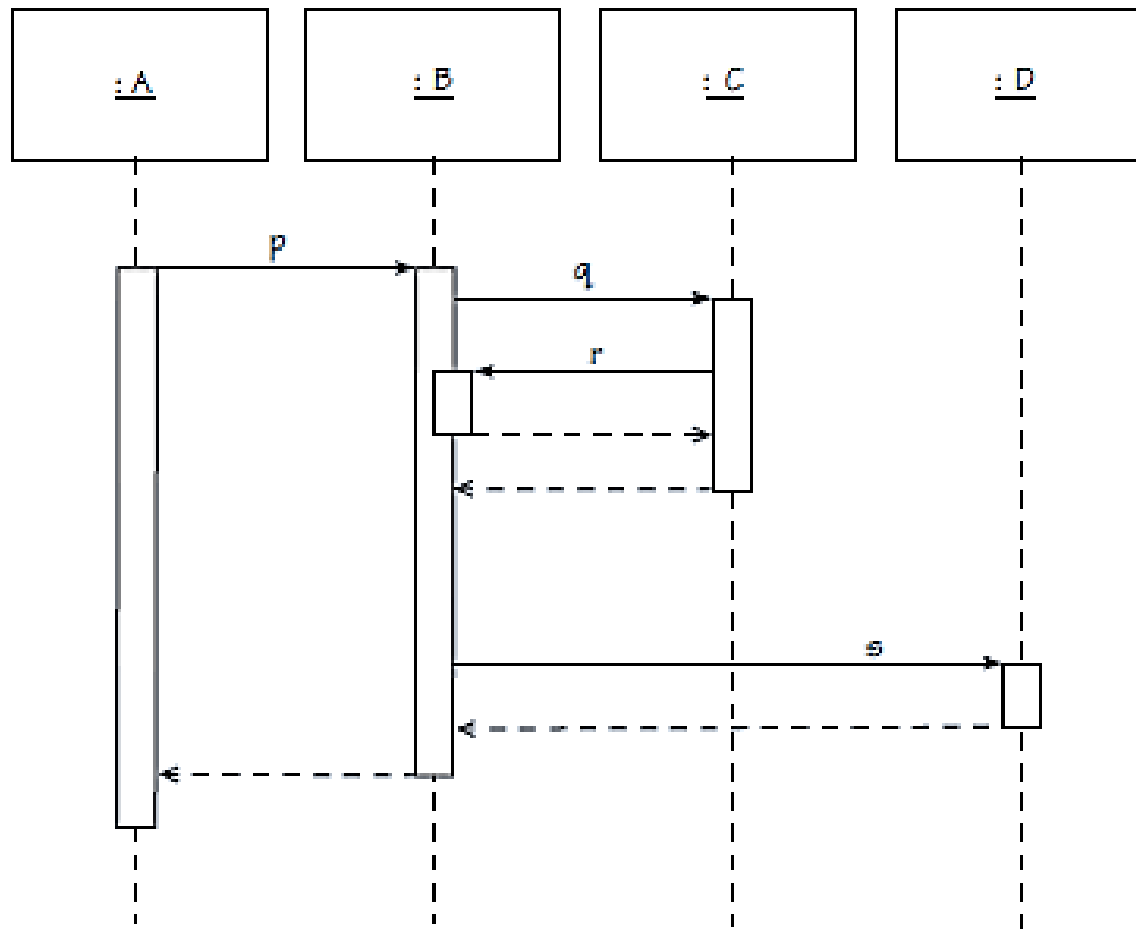
A **communication (collaboration) diagram** is an object diagram with added message-sends.

- Every message has a multi-stage number (1.1).
- The numbers specify the sequencing, replacing vertical position in a sequence diagram.

COLLABORATION DIAGRAMS

COLLABORATION DIAGRAMS

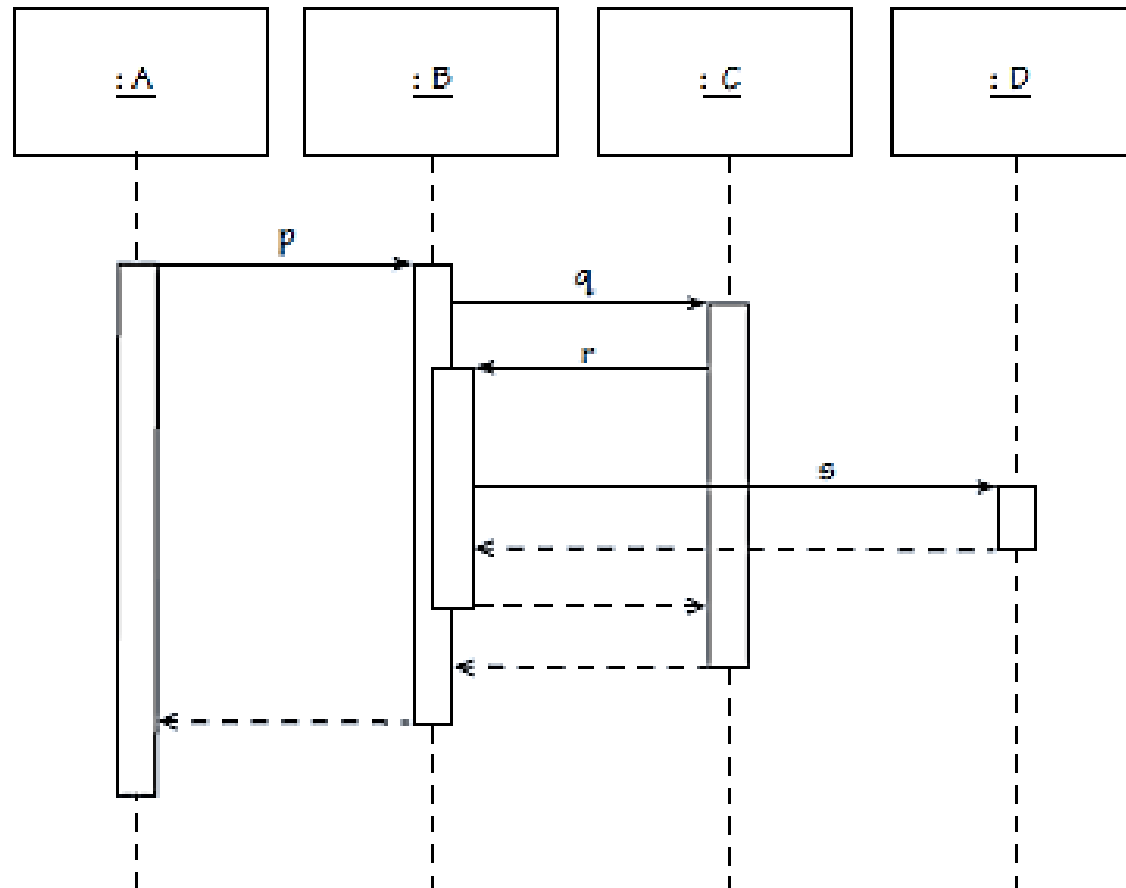
CONVERSION FROM A SEQUENCE DIAGRAM



(a) first message
sequence

COLLABORATION DIAGRAMS

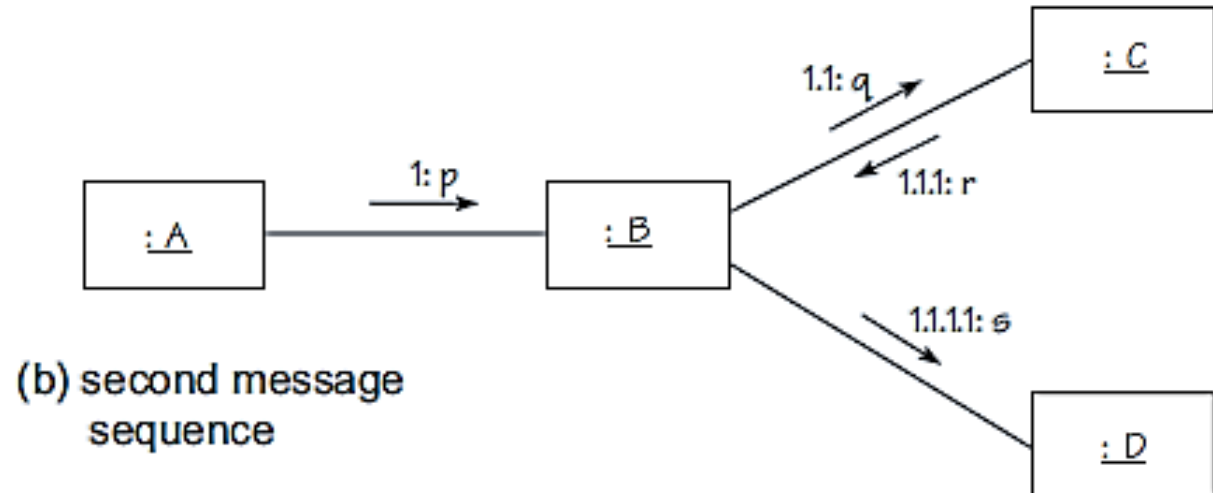
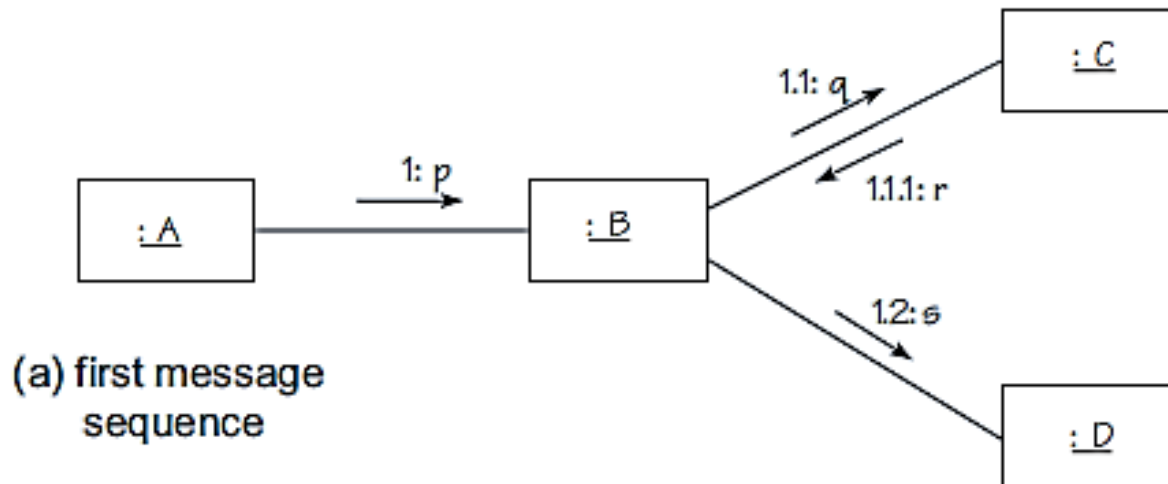
CONVERSION FROM A SEQUENCE DIAGRAM



(b) second message
sequence

COLLABORATION DIAGRAMS

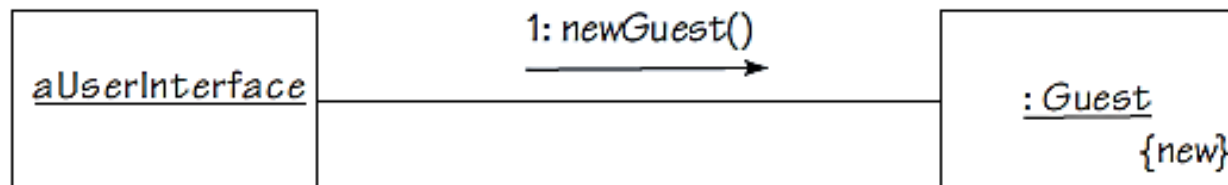
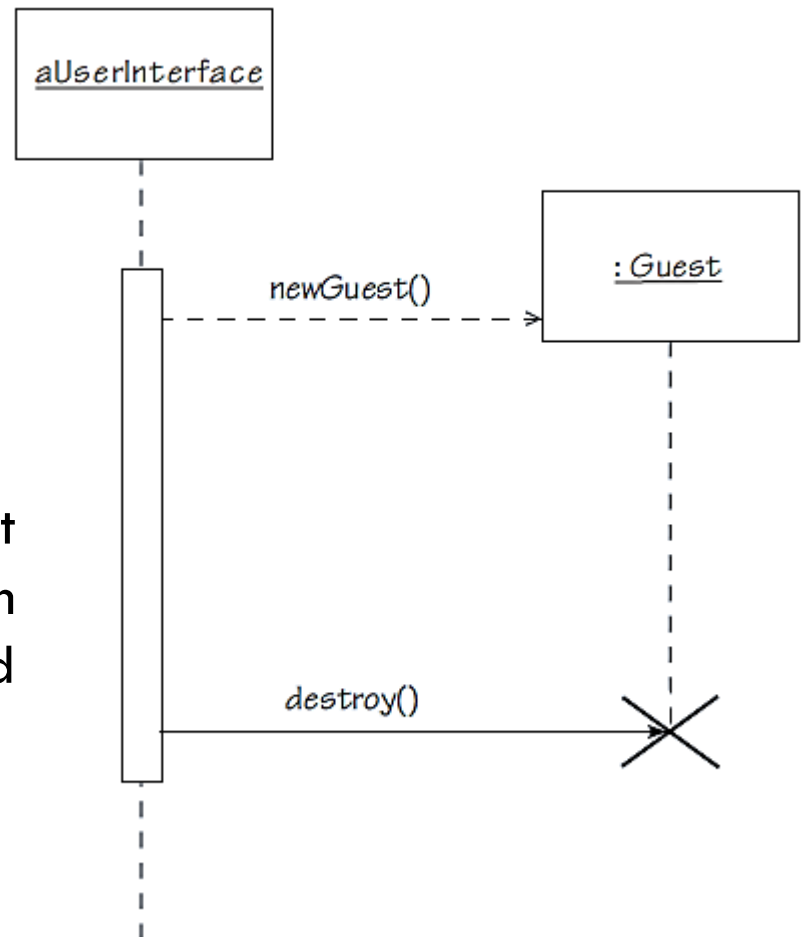
CONVERSION FROM A SEQUENCE DIAGRAM



COLLABORATION DIAGRAMS

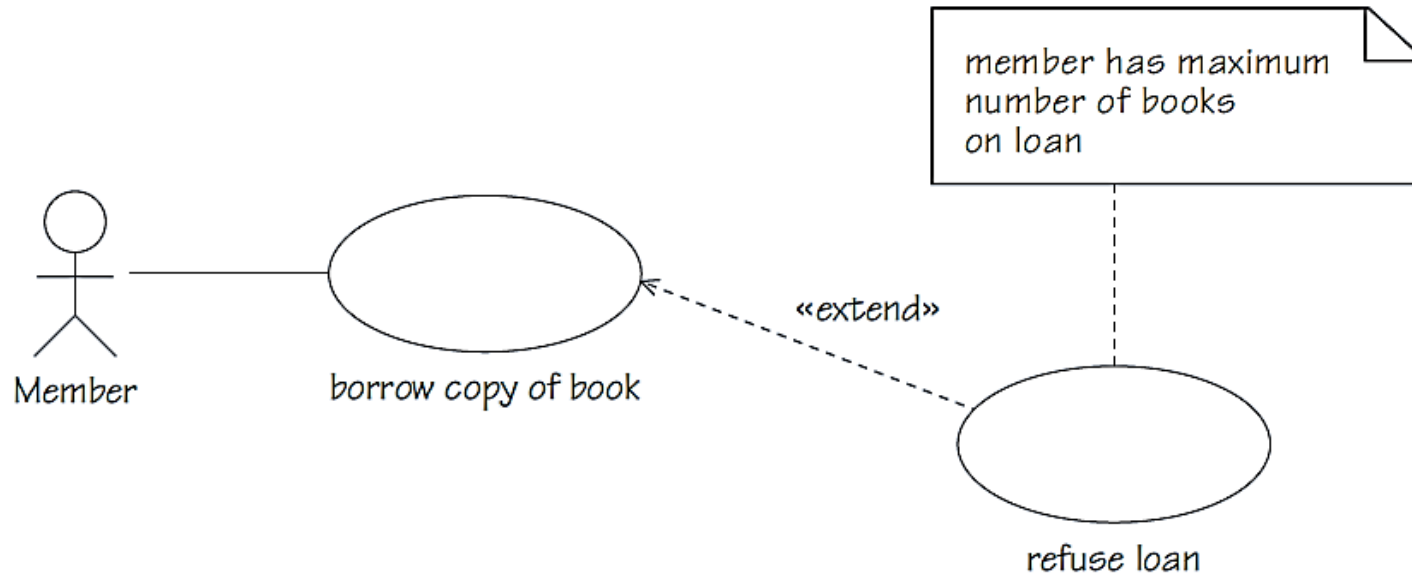
OBJECTS CREATION AND DESTRUCTION

In a communication diagram: object creation and destruction are shown by the special constraints `{new}` and `{destroyed}` respectively.

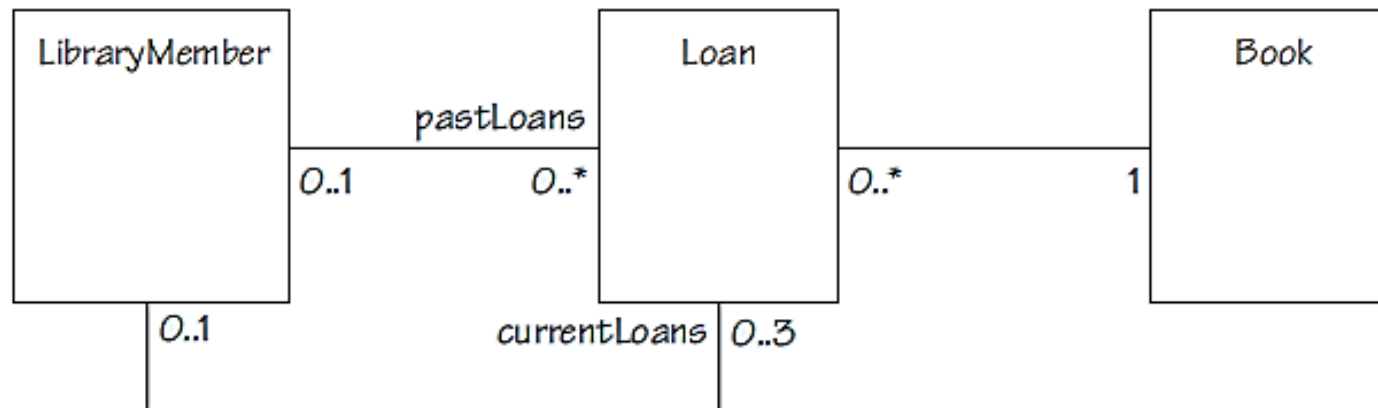


COLLABORATION DIAGRAMS

AN EXAMPLE .. USE-CASE & CLASS DIAGRAMS



► A class model for lending books, where there is a requirement to record both past and current loans



THANK YOU!

Questions?

